

AQA (Trilogy) Combined Science

GCSE Unit 5.4 Chemical Changes

Test (Levels 4 – 9)

Time allowed: **50 minutes**

Question	Links to Student Progress Sheet	Score
1	5.4.1.1 a	
2	5.4.1.2 b, f, g	
3	5.4.1.3 a, b	
4	5.4.1.4 a, b	
5	5.4.2.1 a, 5.4.2.2 b	
6	5.4.2.2 a	
7	5.4.2.3 a, b, c	
8	5.4.2.4 a, d, e	
9	5.4.2.4 b, 5.4.2.5 a, c, d	
10	5.4.3.1 a, 5.4.3.3 a, b	
11	5.4.3.4 a, b	
12	5.4.3.5 a, b, c	
Total	5.4	

Area for Improvement:

Score	Estimated Grade
47 – 50	9
40 – 46	8
33 – 39	7
28 – 32	6
24 – 27	5
20 – 23	4
0 – 19	U

Please be aware that these grade boundaries are Twinkl's own, and may not be in-line with AQA's own.

1. Some metals react with oxygen.

a. Complete the equation below:

Iron + oxygen

[1 mark]

1

2. A student carried out a series of reactions between different metals and sulphate compounds. His results are recorded in the table below. The student put a ✓ to indicate where a reaction had occurred.

	Copper sulphate	Zinc sulphate	Magnesium sulphate
Copper			
Zinc	✓		
Magnesium	✓	✓	

a. List the three metals in order of reactivity, from most reactive to least

reactive. **Most Least**

[2 marks]

b. Complete the equation below:

Copper sulphate + zinc +

[1 mark]

3

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3.

iron

a. Which of the metal(s) below would not need to be extracted from compounds? Place a ✓ in the correct box(es).

aluminium

copper

Chemical Changes **Test [1 mark]**

gold

sodium

iron

aluminium

copper

b. Which of the metal(s) would be extracted by reduction using carbon? Place a ✓ in the correct box(es).

gold

[1 mark] **2**

sodium

4. Sodium chloride is an ionic compound. The reaction is shown in the equation



a. Write the half equation to show the oxidation of sodium.

[2 marks]

b. Write the half equation to show the reduction of chlorine.

[2 marks]

c. What name is give to the type of reaction described above?

[1 mark]

5

5. Complete the word equations below.

a.

[2 marks]

b.

4

[2 marks]

6. Magnesium carbonate reacts with dilute hydrochloric acid.



Some students carried out an investigation of the reaction and collected the gas using a syringe. The apparatus is shown in the diagram below.

gas syringe

conical flask

reaction mixture

a. What gas are the students collecting?

[1 mark]

b. Describe how the students could test for the gas they have collected in a lab. [2 marks]

The diagram below shows the scale on the gas syringe.



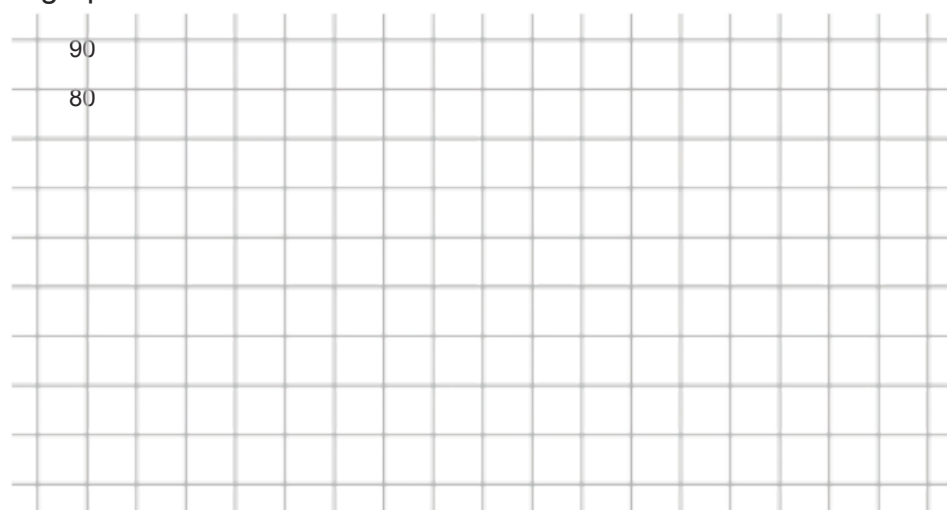
10 20 30 40 50 60 70 80 90 100 C. What is the volume of collected gas in the syringe?

_cm³ [2 marks]

The results of the student’s investigation are shown below.

Mass of magnesium carbonate, g	Volume of gas, cm ³
0.0	0
0.1	15
0.2	32
0.3	46
0.4	58
0.5	71
0.6	76
0.7	78
0.8	79

d. Draw a graph to show the students results. Include a **curve** of best fit.



Volume of gas, cm³

70

60

50

40

30

20

10

0

0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9

Mass of magnesium, g

7

[2 marks]

7. A student is preparing a dry sample of a soluble salt.

a. Below are some of the pieces of equipment available to use:

Conical flask

Filter funnel

Beaker

Copper carbonate powder

Dilute sulphuric acid (0.1M)

Describe a method that the student can use to produce the salt.

You will be assessed on your quality of written communication.

[6

8.

a. Explain the reaction between an acid and a base in terms of ions and pH.

4

[4 marks]

9. Some metals react with dilute acids.

a. What is meant by the term 'dilute'?

[1 mark]

A student tested the acid with universal pH indicator solution and it turned red.

b. What does this tell you about the acid?

[1 mark]

c. Describe the acid in terms of dissociation of ions.

[2 marks]

d. From the information above, suggest the name of an acid the student may

have been using.

[1 mark]

5

10. Aluminium is a metal which is used for the production of food and drinks containers as well as vehicles and other products.

Aluminium must first be extracted from its ore, bauxite, before it can be used.

a. Which method of extraction is used to separate aluminium from its metal ore compound? Place a ✓ in the correct box.

reduction with carbon

electrolysis

reduction by heating alone

[1 mark]

b. Explain why the method chosen above cannot be used on a solid compound.

[2 marks]

3

11. Sodium chloride (table salt) solution can be electrolysed. Two gases are produced.
The gases were collected in test tubes and tested.
Gas A produced a squeaky pop when a lit splint was held into the test tube.
Gas B bleached blue litmus paper placed into the test tube.

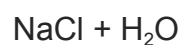
a. Identify the gases produced when sodium chloride is electrolysed.

Gas A:

Gas B:

[2 marks]

Sodium chloride solution is made by adding salt to water. The reactants can be represented by the symbol equation:



b. Circle the ions below which are produced when the water is electrolysed.



[1 mark]

c. Give an example of **one** use for Gas B, identified above.

[1 mark] 4

12. A student is carrying out the electrolysis of lead bromide.

a. What are the substances produced at each of the electrodes?

Positive anode:

Negative cathode:

[2 marks]

b. Complete the half equation for the reduction of

lead at the cathode. $\text{Pb}^{2+} +$

[2 marks] 4

End of Test

